Shahjalal University of Science and Technology
Department of Computer Science and Engineering
4th Year 1st Semester Final Examination, 2024 (Session: 2020-21)
Course Code: CSE 475 Course Title: Machine Learning
Time: 3.0 hrs **Credits: 3.0** **Total Marks: 100**

Group A
[Answer all questions]

1. Answer any FIVE. **5x2=10**

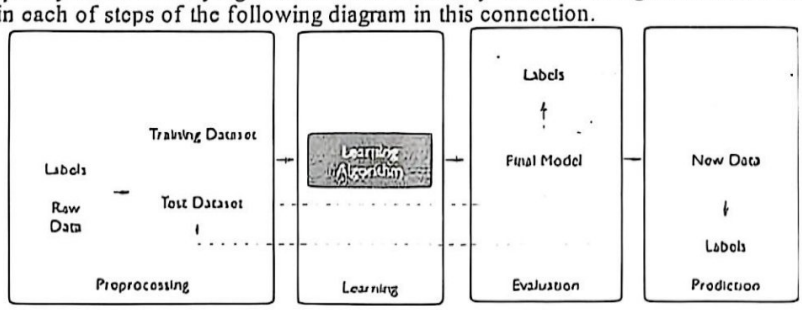
a) What is reinforcement learning?
b) What does the "k" represent in k-NN?
c) Which cost function is used in linear regression? Why?
d) What does "curse of dimensionality" mean?
e) What is the limitation of an MLP model?
f) What is the difference between traditional programming and machine learning?
g) What is the main limitation of ensemble learning?

2. Answer any FOUR. **4x5=20**

a) Define sigmoid and ReLU activation function. Why is ReLU preferred over sigmoid function in neural networks?
b) Explain how L1 regularization performs feature selection.
c) What is the purpose of cross-validation? Explain the underlying process with appropriate diagram.
d) Define the term TF-IDF. How does TF-IDF assess the word relevancy in encoding?
e) What happens when there are outliers in the dataset? Explain the RANSAC algorithm for handling outliers.
f) Why is feature scaling important? Explain the standardization method with an example.

3. Answer any TWO. **2x10=20**

a) i. Differentiate between parametric and non-parametric ML models. Give examples of each type. **3+7**
ii. Suppose you are classifying sentiments of university students on a government's decision. Explain each of steps of the following diagram in this connection.



b) i. Consider the following dataset with features Weather (Sunny, Overcast, Rain) and Temperature (Hot, Mild, Cool) along with the target class (Yes/No). Using the Gini impurity criterion, determine which feature should be chosen for the first split in a decision tree. **6+4**

Instance	Weather	Temperature	Class
1	Sunny	Hot	No
2	Sunny	Hot	No
3	Overcast	Hot	Yes
4	Rain	Mild	Yes
5	Rain	Cool	Yes
6	Rain	Cool	No
7	Overcast	Cool	Yes
8	Sunny	Mild	No

ii. Explain how the boosting approach of an ensemble works?

c) i. Briefly explain the steps for training a neural network. **5+5**
ii. Explain the meaning of the following piece of Python code.

```
model = keras.Sequential([
    layers.Dense(16, activation="relu"),
    layers.Dense(16, activation="relu"),
    layers.Dense(1, activation="sigmoid")
])
```

What is the effect of replacing sigmoid with softmax in a perceptron?

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Group B

[Answer all questions]

4. Answer any FIVE. 5x2 = 10

a) What is the vanishing gradient problem?
b) What is transfer learning?
c) Give one reason why accuracy is not enough for imbalanced datasets.
d) Why should the activation function in ANN be differentiable?
e) Which metric can be used to evaluate a model that outputs a continuous value?
f) What are outliers?
g) What is a hyperparameter? Give suitable examples.

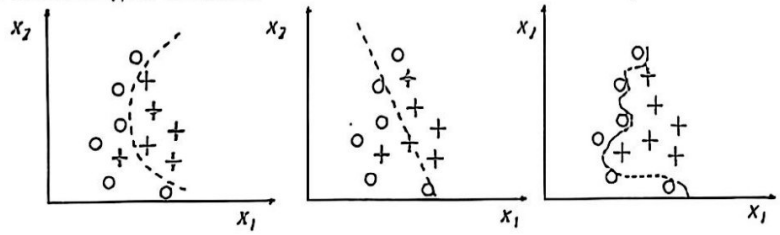
5. Answer any FOUR. 4x5 = 20

a) How is perceptron related with the biological neuron? Given the input vector [1, –2], the weight vector [0.5, 0.25], and a bias of 1, compute the output of a perceptron using a step activation function.
b) Define epoch and batch in the context of ML. If batch size = 128 and dataset has 60000 samples, how many iterations per epoch are required?
c) Define slope, derivative, and gradient? Briefly explain each with figures.
d) What are the limitations of the k-means algorithm? How does the k-means++ algorithm address them?
e) Why is KNN said to be a lazy learner? What happens when the majority voting in KNN results in a tie?
f) What is meant by feature importance? How can it be determined?

6. Answer any TWO. 2x10=20

a) **i.** What does the confusion matrix represent? 3 + 7
ii. A dataset has 100 positives and 50 negatives. A classifier correctly predicts 90 positives and 30 negatives. Compute accuracy, precision, recall, and F1-score. Why is the F1-score useful for imbalanced datasets?

b) **i.** Explain the kernel-trick in SVM to find separating hyperplanes in a high-dimensional space. 5 + 5
ii. Explain which of the following cases represent underfitting, overfitting, and a good fitting. How can overfitting be reduced?



g) Consider your activities of last days: 4+6

Deadline	Weather	Mood	Activity
Urgent	Good	Off	Play
Urgent	Bad	Off	Study
Near	Good	Off	Play
None	Good	Off	Play
None	Bad	On	Sleep
None	Good	Off	Play
Near	Bad	On	Study
Near	Bad	Off	Sleep
Near	Good	Off	Play
Urgent	Bad	On	Study

i. Calculate the prior probabilities: P(Play), P(Study), P(Sleep)
ii. Given that there is a deadline near, the weather is bad and mood is off, find out what you will do today? Use Naive Bayes' classifier

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